

FT 约定与运算规则 (正变换 +2πi !)

- $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{+2\pi i \xi \cdot x} dx, \quad f(x) = \int \widehat{f}(\xi) e^{-2\pi i \xi \cdot x} d\xi$ (与 Stein/Folland 反号, 无 $1/2\pi$)
- 微分→乘法: $\widehat{\partial_x f} = -2\pi i \xi \widehat{f}; \quad \widehat{\Delta f} = -4\pi^2 |\xi|^2 \widehat{f}$
 - 乘法→微分: $\partial_x \widehat{f} = 2\pi i x_j \widehat{f}$
 - 平移: $\widehat{f(\cdot - a)}(\xi) = e^{2\pi i a \xi} \widehat{f}(\xi)$; 调制: $\widehat{e^{2\pi i b x} f}(\xi) = \widehat{f}(\xi + b)$
 - 缩放: $\widehat{f(ax)}(\xi) = \frac{1}{|a|} \widehat{f}(\xi/a)$
 - 卷积: $(f * g)(x) = \int f(x - y) g(y) dy, \widehat{f * g} = \widehat{f} \widehat{g}$; FT 在 $\mathcal{S}_L$ 上单射 (逆推 $u = K * f$ 的依据)

FT 值表 (PF Part I 全部出过)

$f(x)$	$\widehat{f}(\xi)$
$\mathbf{1}_{[-a,a]}$	$\frac{\sin(2\pi a \xi)}{\pi \xi} \quad (\xi \neq 0; \ 2a)$
$\mathbf{1}_{[-a,a] \times [-b,b]}$	张量积: $\frac{\sin(2\pi a \xi_1)}{\pi \xi_1} \cdot \frac{\sin(2\pi b \xi_2)}{\pi \xi_2}$
$e^{-a x }$	$\frac{2a}{a^2 + 4\pi^2 \xi^2}$; 注意 $\frac{1}{1 + 4\pi^2 \xi^2} = \frac{1}{2} e^{- \cdot ^2}$
e^{-ax^2}	$\sqrt{\pi/a} e^{-\pi^2 \xi^2/a}$
$e^{-\pi x^2}$	$e^{-\pi \xi^2}$ (自对偶!)
$e^{-a x ^2} \quad (nD)$	$(\pi/a)^{n/2} e^{-\pi^2 \xi ^2/a}$

Gauss 积分: $\int_{\mathbb{R}} e^{-\pi x^2} dx = 1; \int_{\mathbb{R}} e^{-x^2} dx = \sqrt{\pi}; \int_{\mathbb{R}} e^{-ax^2} dx = \sqrt{\pi/a}$

FT 解 PDE 模板 (PF Part III.1/III.2)

通用算法: FT 两边 → 带系数算子变乘法 (symbol) → 解出 $\widehat{u}$ → 识别核 + 卷积定理 + 单射性逆推。

热方程 $u_t = u_{xx}, u(x, 0) = f; \partial_t \widehat{u} = -4\pi^2 \xi^2 \widehat{u} \rightarrow \widehat{u} = \widehat{f} e^{-4\pi^2 \xi^2 t} \rightarrow$ 识别 $G_t(x) = \frac{e^{-\frac{x^2}{4t}}}{\sqrt{4\pi t}}, \widehat{G_t} = e^{-4\pi^2 t \xi^2} \rightarrow u(x, t) = (G_t * f)(x) = \frac{1}{\sqrt{4\pi t}} \int e^{-(x-y)^2/4t} f(y) dy \quad (nD: (4\pi t)^{-n/2})$

Poisson 型 $-u'' + u = f; (1 + 4\pi^2 \xi^2) \widehat{u} = \widehat{f} \rightarrow u = \frac{1}{2} e^{-|\cdot|} * f = \frac{1}{2} \int e^{-|x-y|} f(y) dy$

$\mathbb{R}^3$ Newtonian: $-\Delta u = f \rightarrow u = \frac{1}{4\pi} \int \frac{f(y)}{|x-y|} dy$; Yukawa $-\Delta u + m^2 u = f \rightarrow$ 核 $\frac{e^{-m|x|}}{4\pi|x|}$

Laplace 值表与规则

- $\mathcal{L}(f)(s) = \int_0^\infty f(t) e^{-st} dt$
- $\mathcal{L}(f') = s\mathcal{L}(f) - f(0); \mathcal{L}(f'') = s^2\mathcal{L}(f) - sf(0) - f'(0)$ **初值勿漏!**
 - $\mathcal{L}(1) = \frac{1}{s}; \mathcal{L}(t^m) = \frac{m!}{s^{m+1}}; \mathcal{L}(e^{-at}) = \frac{1}{s+a}; \mathcal{L}(t^m e^{-at}) = \frac{m!}{(s+a)^{m+1}}$
 - $\mathcal{L}(\sin bt) = \frac{b}{s^2 + b^2}; \mathcal{L}(\cos bt) = \frac{s}{s^2 + b^2}; \mathcal{L}(\sinh bt) = \frac{b}{s^2 - b^2}; \mathcal{L}(\cosh bt) = \frac{s}{s^2 - b^2}$
 - 频移: $\mathcal{L}(e^{-at} f(t))(s) = \mathcal{L}(f)(s + a)$
 - s -求导: $\mathcal{L}(t f(t)) = -\frac{d}{ds} \mathcal{L}(f) \Rightarrow \mathcal{L}(t \sin t) = \frac{2s}{(s^2 + 1)^2}, \mathcal{L}(t \cos t) = \frac{s^2 - 1}{(s^2 + 1)^2}, \frac{1}{(s^2 + 1)^2} = \mathcal{L}(\frac{1}{2}(\sin t - t \cos t))$
 - 卷积: $(f * g)(t) = \int_0^t f(t - y) g(y) dy, \mathcal{L}(f * g) = \mathcal{L}(f) \mathcal{L}(g)$ (积分限 $[0, t]$!)

LT 解 ODE 四步 + PF 两例 (III.3/III.4)

- ① 取 $\mathcal{L}$ 带初值 ② 代数解出 $Y(s)$ ③ 部分分式 ④ 查表逆变换; 末尾代回 $y(0), y'(0)$ 验算
- $y' + 3y = e^{-t}, y(0) = 2: Y = \frac{2s+3}{(s+1)(s+3)} = \frac{1/2}{s+1} + \frac{3/2}{s+3} \rightarrow y = \frac{1}{2} e^{-t} + \frac{3}{2} e^{-3t}$
 - $y'' + y = \sin t, y(0) = 0, y'(0) = 1$ (共振): $Y = \frac{1}{s^4 + 1} + \frac{1}{(s^2 + 1)^2} \rightarrow y = \frac{3}{2} \sin t - \frac{1}{2} \cos t$

部分分式速算: $\frac{1}{(s+1)(s+3)} = \frac{1/2}{s+1} - \frac{1/2}{s+3}$ (盖住法: 在极点处求值)

收敛定理全家 (black box 引用)

- MCT** $\uparrow$ (4.11.18): $0 \leq f_k \uparrow$ a.e., $\sup_k \int f_k < \infty \Rightarrow f := \lim f_k \in \mathcal{S}_L, \int f = \lim \int f_k$. **MCT** $\downarrow$: $f_k \downarrow f$ a.e. + 某 $\int f_k$ 有限 (证: $f_1 - f_k$ 翻转)
- DCT** (4.11.19): $f_k \rightarrow f$ a.e. + $|f_k| \leq_L F \in \mathcal{S}_L \Rightarrow f \in \mathcal{S}_L, \int |f_k - f| \rightarrow 0, \int f_k \rightarrow \int f$. **必须验 $\int F < \infty$**
- Fatou (★ 坎点)**: $f_k \geq 0, \liminf \int f_k < \infty \Rightarrow \liminf f_k \in \mathcal{S}_L$ 且 $\int \liminf f_k \leq \liminf \int f_k$.
证明骨架: $g_k := \inf_{j \geq k} f_j; (i) \ 0 \leq g_k \leq f_k \Rightarrow g_k \in \mathcal{S}_L$ (ii) g_k 单调增 (iii) $\lim g_k = \liminf f_k; \int g_k \leq \inf_{j \geq k} \int f_j$, 套递增 MCT.
- Fatou a.e. 版**: $f_k \geq 0, f_k \rightarrow f$ a.e. $\Rightarrow \int f \leq \liminf \int f_k$ (衔接句: a.e. 收敛 $\Rightarrow \liminf f_k = f$ a.e.)
- Reverse Fatou**: 另设 $f_k \leq F \in \mathcal{S}_L \Rightarrow \limsup \int f_k \leq \int \limsup f_k$ (对 $F - f_k$ 用 Fatou, 消有限 $\int F$)
- Gen. DCT**: $|f_k| \leq g_k, g_k \rightarrow g$ a.e., $\int g_k \rightarrow \int g < \infty, f_k \rightarrow f$ a.e. $\Rightarrow \int f_k \rightarrow \int f$ (Fatou 于 $g_k \pm f_k$ 双侧夹; "收敛项可提出 $\liminf$ ")
- Scheffé**: $f_k \rightarrow f$ a.e. $\Rightarrow [\int |f_k| \rightarrow \int |f| \iff \int |f_k - f| \rightarrow 0]$ (取 $h_k = |f_k - f| \leq |f_k| + |f| \Rightarrow g_k \rightarrow 2|f|$, 套 Gen. DCT)
- 可积性自适模式: $f_k \geq 0, f_k \rightarrow f$ a.e., $\sup \int f_k = C < \infty \Rightarrow f \in \mathcal{S}_L$ 且 $\int f \leq C$ (Fatou, 不需要 dominator!)
- 桥定理 (4.11.19 题): 非负 $f: f \in \mathcal{S}_L \iff$ 改进 Riemann 积分绝对收敛, 且数值相等。反例 $\frac{\sin x}{x}$: 条件收敛 (分部积分) 但 $\notin \mathcal{S}_L$ ($|\cdot|$ 逐段 $\geq \frac{2}{(n+1)\pi}$, 调和发散)

决策树 + 答题模板 + 反例

选定理: 单调? $\rightarrow$ MCT | 固定可积 dominator? $\rightarrow$ DCT | 非负 + 单边不等式? $\rightarrow$ Fatou | dominator 在动? $\rightarrow$ Gen DCT | 问 $\int |f_k - f| \rightarrow 0$? $\rightarrow$ Scheffé

模板 (4–10 行): 点名定理 $\rightarrow$ 逐条验证 (单调性 / $\int F < \infty$ 算出来 / a.e. 极限求出来) $\rightarrow$ 套结论

- Traveling box $\mathbf{1}_{[k,k+1]}$** : 逃 ∞ ; dominator $\geq \mathbf{1}_{[1,\infty)}$ 不可积
- Spike $k \mathbf{1}_{(0,1/k]} / \frac{k}{1+k^2} \mathbf{1}_{[0,1]}$** ($\int = \arctan k \rightarrow \frac{\pi}{2}$): 集中到点; dominator $\gtrsim \frac{1}{x}$ 不可积
- 震荡箱 ($\mathbf{1}_{[0,1]} / \mathbf{1}_{[1,2]}$ 交替): $\liminf$ 逐点取最差子列, $\int \liminf = 0 < 1$ — **一致有界 + 公共支撑也救不了, 还缺逐点收敛**
- 负 **spike** $-k \mathbf{1}_{(0,1/k]}$: 非负假设删掉 Fatou 反向

径向积分 & CoV Jacobian 工具箱

- 极坐标 | $\det| = r$; 柱坐标 = r ; 球坐标 (纬度 $\varphi, z = r \sin \varphi) = r^2 \cos \varphi$
- 径向降维: $\int_{\mathbb{R}^n} g(|\mathbf{x}|) dV = \sigma_n \int_0^\infty g(r) r^{n-1} dr; \sigma_1 = 2, \sigma_2 = 2\pi, \sigma_3 = 4\pi, \sigma_n = n \rho_n$ (单位球面积; 有限常数, 不影响收敛性判断)
- p -阈值: $\int_{|\mathbf{x}| \geq 1} |\mathbf{x}|^p < \infty \iff p < -n; \int_{|\mathbf{x}| \leq 1} |\mathbf{x}|^p < \infty \iff p > -n$ (球壳 r^{n-1} 抬指数)
- 1-D 判据: $\int_0^1 x^a < \infty \iff a > -1; \int_1^\infty x^a < \infty \iff a < -1; \int_0^1 (-\ln x) = 1$
- 夹逼: 证可积只要上界 $f \leq Cg$; 证发散只要下界 $f \geq cg$; 双边夹 $\Rightarrow$ 同收敛
- 指数压多项式: $e^t \geq t^k/k! \Rightarrow e^{-t^2} \leq k! t^{-2k}$ (k 任选) $\Rightarrow \int r^N e^{-r^2} dr < \infty$, Gaussian 矩全存在

平面曲线 $c(t) = (x, y)$

- $\kappa = \frac{x'y'' - y'x''}{((x')^2 + (y')^2)^{3/2}} = \frac{\det(\dot{c}, \ddot{c})}{|\dot{c}|^3}$ (signed, + = 向左弯)
- 图象 $(t, f(t))$: $\kappa = \frac{f''}{(1 + (f')^2)^{3/2}}$; 圆 (半径 a): $\kappa \equiv 1/a$

PF Part IV	$\kappa(t)$	看点
(t, t^2)	$\frac{2}{(1 + 4t^2)^{3/2}}$	顶点 $\kappa(0)=2$
$(a \cos t, a \sin t)$	$1/a$	常数
(t, t^3)	$\frac{6 t }{(1 + 9t^4)^{3/2}}$	$\kappa(0)=0$ 拐点
(t, e^t)	$\frac{e^t}{(1 + e^{2t})^{3/2}}$	$t \rightarrow -\infty$ 趋直线

空间曲线: 公式 + 流程

- $\kappa = \frac{|\dot{c} \times \ddot{c}|}{|\dot{c}|^3} > 0$; $\tau = \frac{\det(\dot{c}, \ddot{c}, \dddot{c})}{|\dot{c} \times \ddot{c}|^2}$ **τ 分母 = 叉积平方, 不是 $|\dot{c}|^6$!**
- 流程: 算 $\dot{c}, \ddot{c}, \dddot{c} \rightarrow \dot{c} \times \ddot{c}$ 与 $|\dot{c} \times \ddot{c}| \rightarrow \kappa \rightarrow \det[\dot{c}|\ddot{c}|\dddot{c}] \rightarrow \tau$ 。 **det 技巧**: 多项式曲线 (t, t^2, t^3) 型的 $[\dot{c}|\ddot{c}|\dddot{c}]$ 是上三角, det = 对角积; 螺旋型沿"常数列"展开
- 不需要先化 unit-speed (公式自带 $|\dot{c}|$); 直线 $\ddot{c} = 0$: Frenet 失效, τ 无定义
- Frenet: $\mathbf{T} = \frac{\dot{c}}{|\dot{c}|}$, $\mathbf{N} = \frac{\dot{c} \times \ddot{c}}{|\dot{c} \times \ddot{c}|}$, $\mathbf{B} = \mathbf{T} \times \mathbf{N}$; unit-speed: $\mathbf{T}' = \kappa \mathbf{N}$, $\mathbf{N}' = -\kappa \mathbf{T} + \tau \mathbf{B}$, $\mathbf{B}' = -\tau \mathbf{N}$ (OH: frame 理论大概率不考)

PF Part V	κ	τ
$(\cos t, \sin t, t)$	$1/2$	$1/2$
$(2 \cos t, 2 \sin t, t)$	$2/5$	$1/5$
$(t, t^2, 0)$	$\frac{2}{(1 + 4t^2)^{3/2}}$	0 (平面曲线)
(t, t^2, t^3)	$\frac{2 \sqrt{9t^4 + 9t^2 + 1}}{(1 + 4t^2 + 9t^4)^{3/2}}$	$\frac{3}{9t^2 + 9t^2 + 1}$

一般螺旋 $(a \cos t, a \sin t, bt)$: $\kappa = \frac{a}{a^2 + b^2}$, $\tau = \frac{b}{a^2 + b^2}$ (验算神器); $\tau \equiv 0 \iff$ 曲线落在固定平面内

曲面六步流水线

- ① $X_u, X_v \rightarrow \mathbf{I} = \begin{pmatrix} E & F \\ F & G \end{pmatrix}$, $E = X_u \cdot X_u$, $F = X_u \cdot X_v$, $G = X_v \cdot X_v$
- ② $\nu = \frac{X_u \times X_v}{|X_u \times X_v|}$, $|X_u \times X_v| = \sqrt{EG - F^2}$; 面积 = $\iint \sqrt{EG - F^2}$ **不是 $\sqrt{EG}$**
- ③ $X_{uu}, X_{uv}, X_{vv} \rightarrow \mathbf{II} = \begin{pmatrix} L & M \\ M & N \end{pmatrix}$, $L = X_{uu} \cdot \nu$ 等
- ④ $K = \frac{LN - M^2}{EG - F^2}$, $H = \frac{EN - 2FM + GL}{2(EG - F^2)}$ **分母有 2**
- ⑤ $S = \mathbf{I}^{-1}\mathbf{II}$, 主曲率 = 特征值; $K = k_1 k_2$, $H = \frac{k_1 + k_2}{2}$; 由 H, K 反解: $k_{1,2} = H \pm \sqrt{H^2 - K}$
- ⑥ $F = M = 0$ 时直接 $k_1 = L/E$, $k_2 = N/G$ (球/柱/旋转面/torus 全走这条)
- K 内蕴 (Theorema Egregium), 不依赖 ν ; H 随 ν 翻转变号; $K=0 \neq$ 平面 (柱/锥 developable)

Graph $z = f(x, y)$, $W = \sqrt{1 + f_x^2 + f_y^2}$

- $\mathbf{I} = \begin{pmatrix} 1 + f_x^2 & f_x f_y \\ f_x f_y & 1 + f_y^2 \end{pmatrix}$, $\nu = \frac{(-f_x, -f_y, 1)}{W}$, $\mathbf{II} = \frac{1}{W} \begin{pmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{pmatrix}$
- $K = \frac{f_{xx}f_{yy} - f_{xy}^2}{W^4}$, $H = \frac{(1 + f_y^2)f_{xx} - 2f_x f_y f_{xy} + (1 + f_x^2)f_{yy}}{2W^3}$
- PF VII.4 椭圆抛物面 $z = \frac{x^2 + y^2}{2}$ ($r^2 = x^2 + y^2$, $w = 1 + r^2$): $K = \frac{1}{w^3}$, $H = \frac{2 + r^2}{2w^2}$, $k_1 = \frac{1}{\sqrt{w}}$, $k_2 = \frac{1}{w^{3/2}}$
- 双曲抛物面 $z = \frac{x^2 - y^2}{2}$: $K = \frac{-1}{w^3}$, $H = \frac{y^2 - x^2}{2w^{5/2}}$, minimal points: $y = \pm x$

标准曲面结果表 (PF Part VII + 冲刺页)

曲面	k_1, k_2	K	H
平面	$0, 0$	0	0
圆柱 R (外法向)	$-\frac{1}{R}, 0$	0	$-\frac{1}{2R}$
球面 R (内法向)	$\frac{1}{R}, \frac{1}{R}$ 脐点	$\frac{1}{R^2}$	$\frac{1}{R}$
鞍面 $z=uv$ @ 0	± 1	-1	0 (minimal)
旋转面 $(r(u) \cos v, r(u) \sin v, h(u))$	$\mathbf{I} = \text{diag}((r')^2 + (h')^2, r^2), F = 0$		

K 与 ν 方向无关; k_i, H 的符号随 ν 翻转 — 报 $|H|$ 或注明法向。 K 符号分类: > 0 椭圆型 / $= 0$ 抛物型 / < 0 双曲(鞍)型

Torus (PF Part VIII, Final Review 压轴)

$X = ((R + r \cos v) \cos u, (R + r \cos v) \sin u, r \sin v)$, $R > r > 0$; $v=0$ 最外圈, $v=\pi$ 最内圈

- $\mathbf{I} = \text{diag}((R + r \cos v)^2, r^2)$; 外法向 $\nu = (\cos v \cos u, \cos v \sin u, \sin v)$; $\mathbf{II} = \text{diag}(-(R + r \cos v) \cos v, -r)$
- $k_1 = \frac{-\cos v}{R + r \cos v}$, $k_2 = -\frac{1}{r}$; $K = \frac{\cos v}{r(R + r \cos v)}$, $H = -\frac{R + 2r \cos v}{2r(R + r \cos v)}$
- $\text{sgn } K = \text{sgn}(\cos v)$: 外半圈 ($|v| < \frac{\pi}{2}$) $K > 0$ / 内半圈 $K < 0$ / 顶底圈 ($v = \pm \frac{\pi}{2}$) $K = 0$
- Minimal ($H=0 \iff \cos v = -\frac{R}{r}$): $R > 2r$ 无 | $R = 2r$: $v=\pi$ 最内圈 | $r < R < 2r$: 两圆周 $v = \pm \arccos(-\frac{R}{r})$

考场提醒 (低级错误清单)

- FT 约定: 本课正变换 $e^{+2\pi i \zeta x}$ (与多数教材反!); 微分公式 $-2\pi i \xi$
- $\mathcal{L}(f')$ 漏 $-f(0) =$ 全错; ODE 答案代回初值验算 (10 秒保命)
- τ 分母 $|\dot{c} \times \ddot{c}|^2$; 平面 κ signed / 空间 $\kappa > 0$
- K 用 W^4 , H 用 $2W^3$ 且分子交叉配权 ($1 + f_y^2$ 配 f_{xx} !); 面积元 $\sqrt{EG - F^2}$
- Fatou 方向: $\int \liminf \leq \liminf \int$ (质量只漏不涨); 非负 + $\liminf \int < \infty$ 两个假设都要写
- DCT 的 dominator 可积性是要验的: 算出或界出 $\int F < \infty$
- 球坐标 Jacobian $r^2 \cos \varphi$ (纬度约定); 热核里 $a = \frac{1}{4t}$ 代 Gaussian 公式别代错